ME 282

Tensile Test



Tension Test

- Test Specimen : Dog-bone specimen
- Some Definitions:

$\sigma_e = P/A_0$ (1)

$\varepsilon_e = \Delta L/L_0$ (2)

Where

σ_e = Eng. stress

ε_e = Eng. strain

P= applied load;

A_0 = initial (original cross-sectional area)

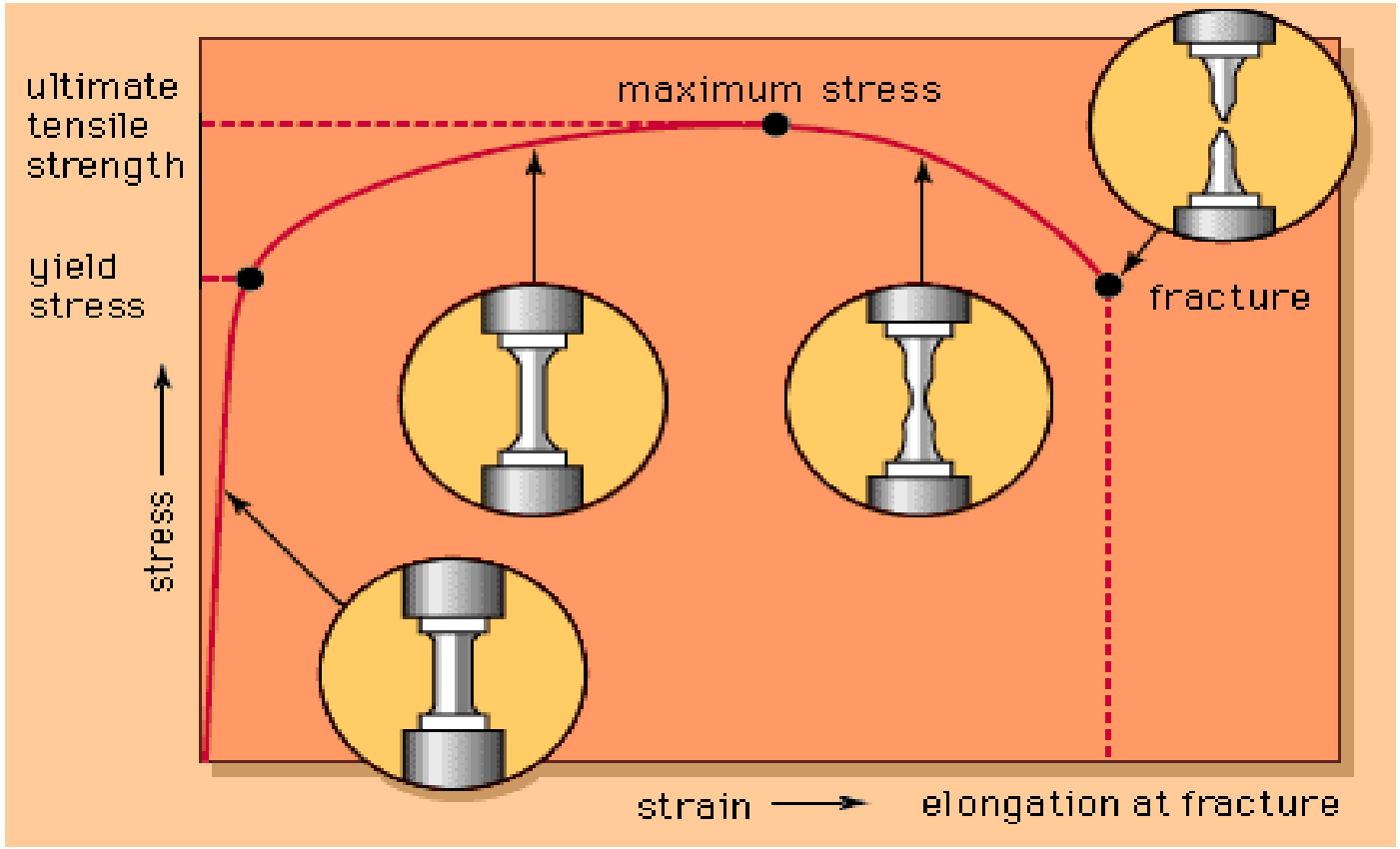
L_0 = initial (original Gauge length)

ΔL = change in length= $L_n - L_0$; (Ln = new /instantanous length)

Test procedure

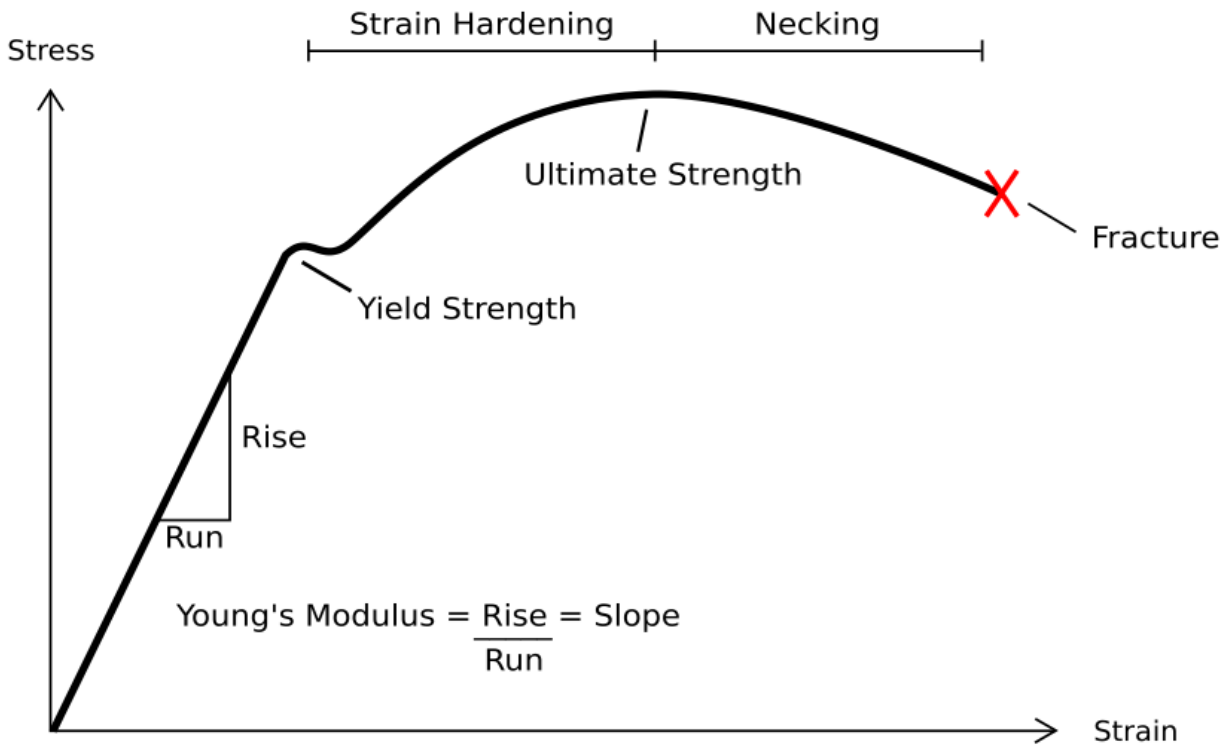
1. Measure specimen diameter or width and depth (necessary to determine A_0)
2. Measure gauge length (L_0)
3. Put specimen in **grips** of tensile machine
4. Pull specimen to failure (fracture)
5. Measure load (P) and new length (L) of specimen during the test.
6. Determine stresses and strains from loads and new length data
7. Plot graph of stress versus strain

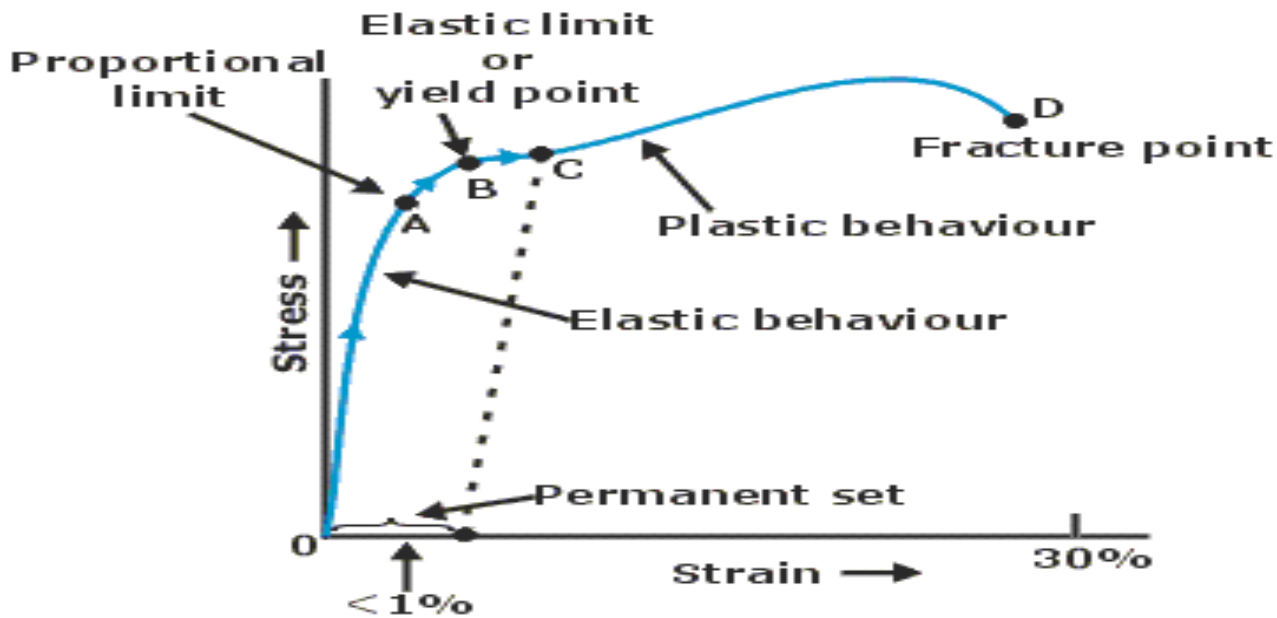
Stress-strain characteristics

- 

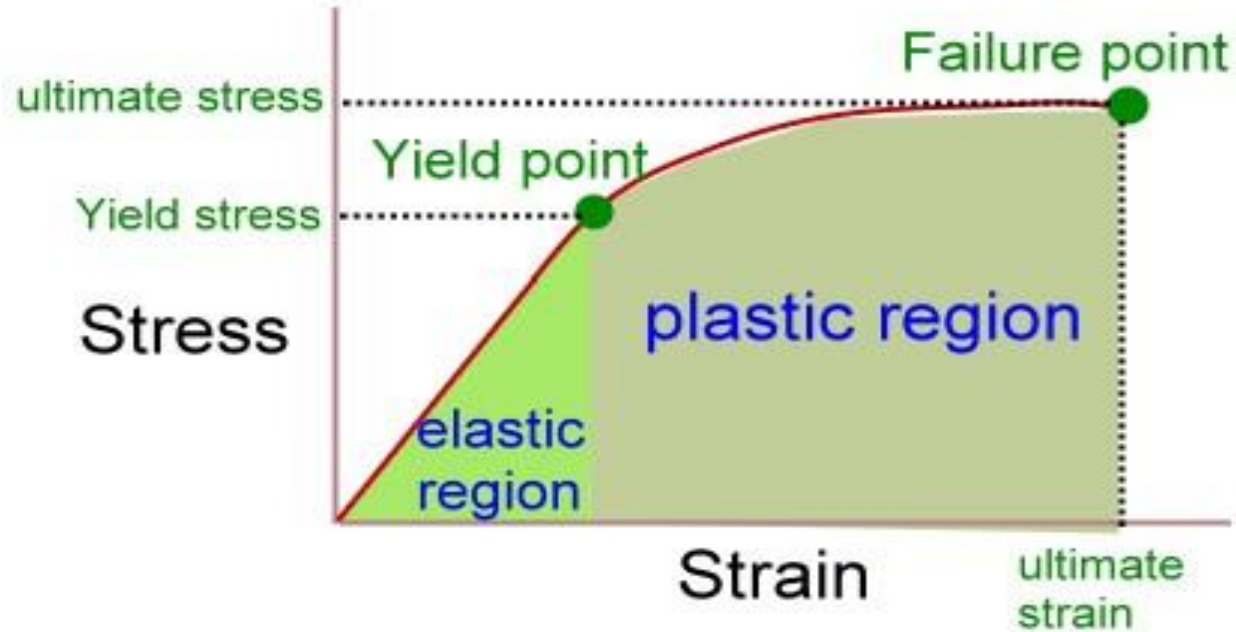
Stress-strain curve

- Typical Engineering Material (Ductile)





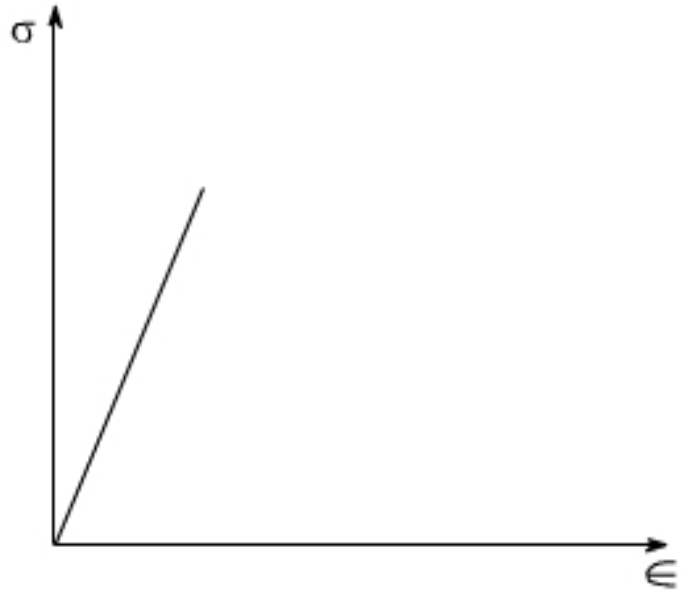
A typical stress-strain curve for a ductile metal


Failure point
ultimate stress
Yield point
Yield stress
Stress
plastic region
elastic
region
Strain
ultimate
strain

Tensile stress-strain curve

- Linear Elastic Behaviour
- Brittle materials:

(1) Glasses
(2) Many Ceramics
(3) Rocks and stones
(4) Heavily cross-linked polymers
(5) Some metals at low temps.



Linearly elastic material

Determination of material properties

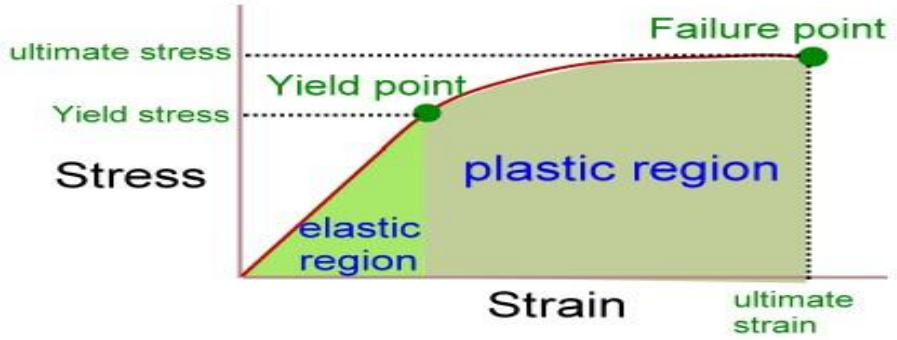
- **Yield strength** σ_y = Yield load ÷ initial X-section area
- **Tensile strength** σ_u = maximum load ÷ initial X-section area (or maximum stress material can withstand before fracture)
- **Modulus of Elasticity** (or Young's Modulus), E:
- $E = \Delta\sigma \div \Delta\varepsilon$ (within elastic region) (3)
- **Resilience**, U_R = Area under the elastic region.
- It is the energy that can be absorbed under elastic loading and completely recovered when load is removed
- $U_R = (1/2)(\sigma_y)^2/E$

Resilience is useful for mechanical springs as in shock absorbers; leaf springs, helical springs, etc.

Material Properties

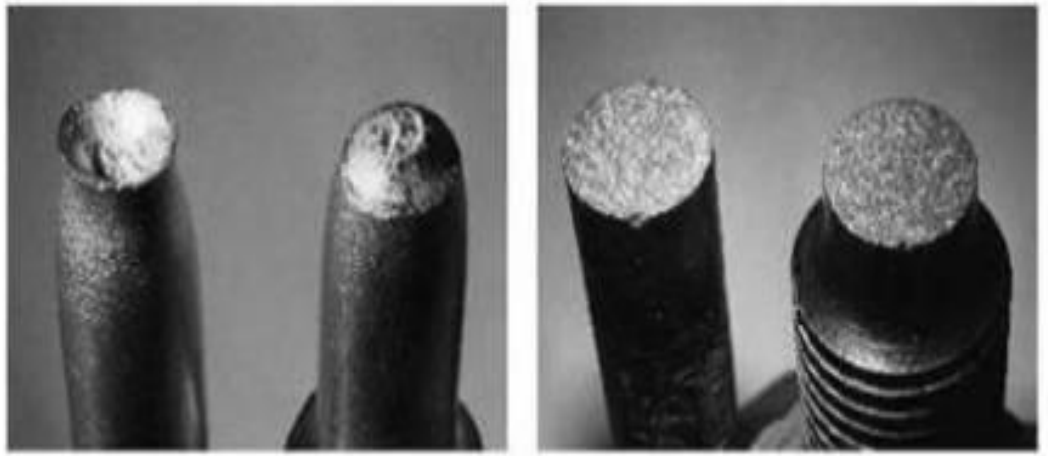
- **Toughness** U = Area under complete stress-strain curve

  It is the total energy a material can absorbed before fracture



- **Percentage elongation** (%El) is the total strain a material withstands before fracture
- It is a measure of ductility (or brittleness) of a material

Ductile and brittle fractures (metals)



True stress-strain curve

Definitions:

- **True Stress** (σ_t) = P/A_i (4)

Where A_i is instantaneous area

- **True Strain** (ε_t) = $\ln (l_n/l_0)$ (5)

l_n= new or instantaneous length

l_0= initial or original length

Assuming constant volume:

$\sigma_t = \sigma_e(1+\varepsilon_e)$ (6)

$\varepsilon_t = \ln(1+ \varepsilon_e)$ (7)

Stress-strain Models

- **Elastic region:**

$\sigma_e = E\varepsilon_e$ (8)

Plastic regime

$\sigma_t = K(\varepsilon_p)^n$ (9)

Where K= stress coefficient;

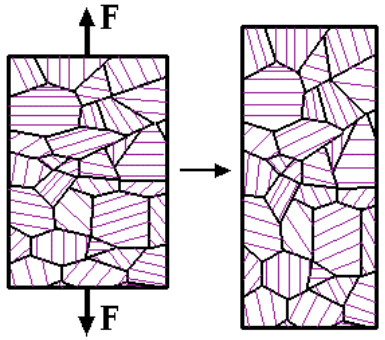
ε_p= true plastic strain;

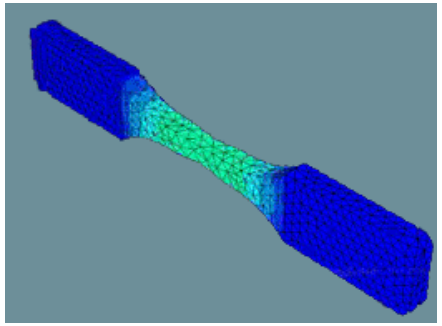
n=strain hardening exponent

Deformation

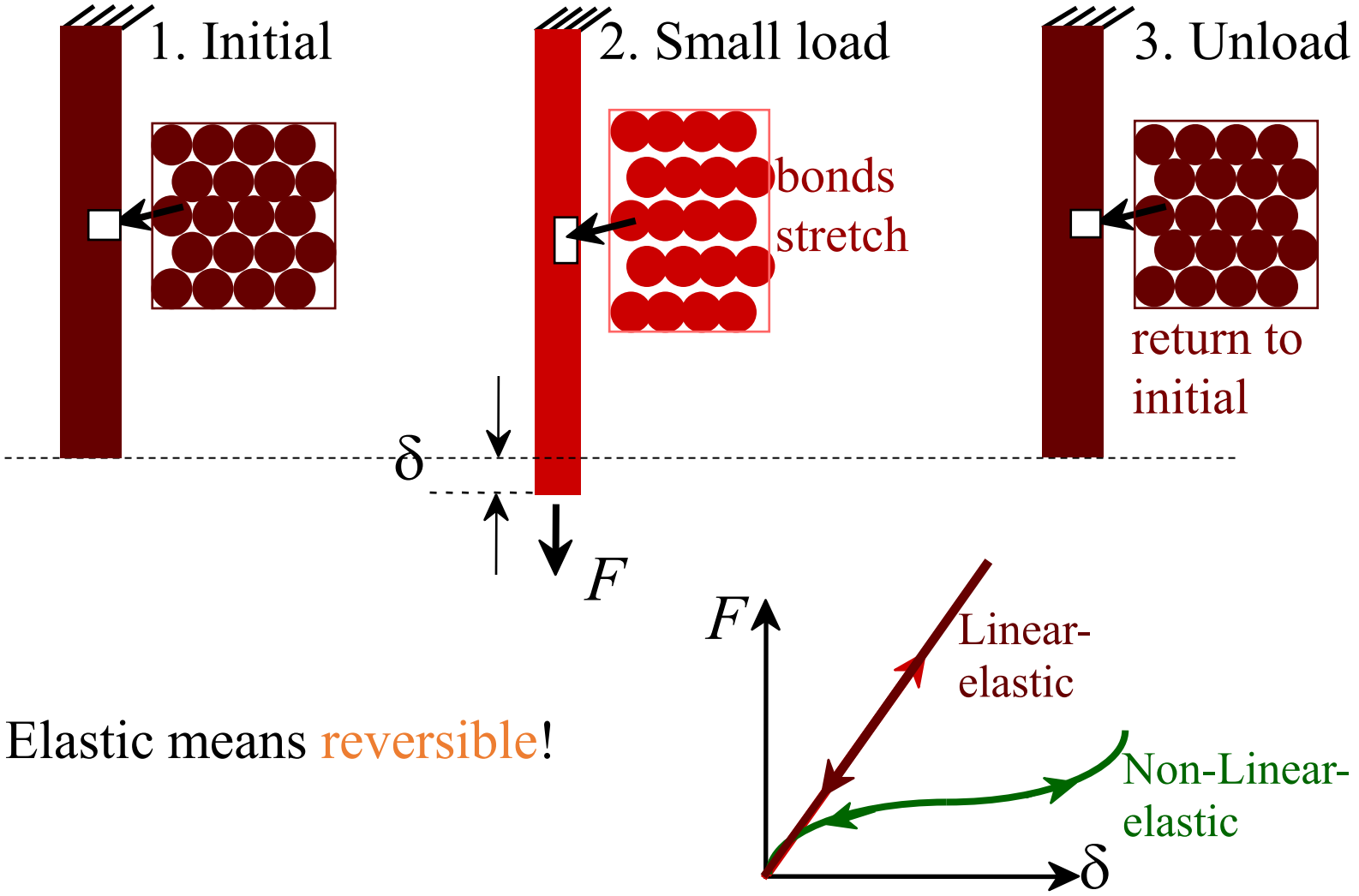
- **elasticity**
  - ability to return to original shape after being deformed by stress
  - *rubber ball or piece of elastic*
- **plasticity**
  - retains new shape after being deformed by stress
  - *wet clay ball or piece of saran wrap*







Elastic Deformation



Elastic means reversible!

Plastic Deformation

- Permanent plastic deformation is due to shear process–atoms change their neighbors.
- Inter-atomic forces and crystal structure plays an important role during plastic deformation.
- Cumulative movement of dislocations leads to gross plastic deformation.
- Edge dislocation move by slip and climb, while screw dislocation move by slip and cross-slip.
- During their movement, dislocations tend to interact. The interaction is very complex because of number of dislocations moving over many slip systems in different directions

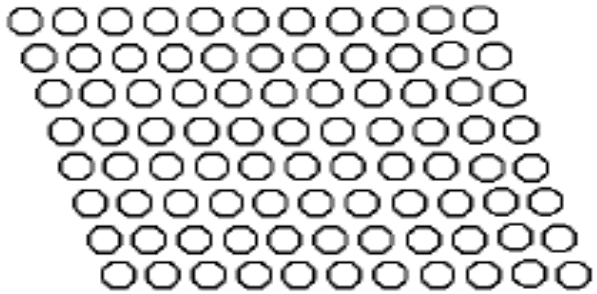
Mechanisms of Plastic deformation -Slip

- Mainly two kind: slip and twinning.
- Slip is prominent among the two. It involves sliding of blocks of crystal over other along slip planes.
- Slip occurs when shear stress applied exceeds a critical value.
- Slip occurs most readily in specific directions(slip directions) on certain crystallographic planes.
- Feasible combination of a slip plane together with a slip direction is considered as a slip system.
- During slip each atom usually moves same integral number of atomic distances along the slip plane.

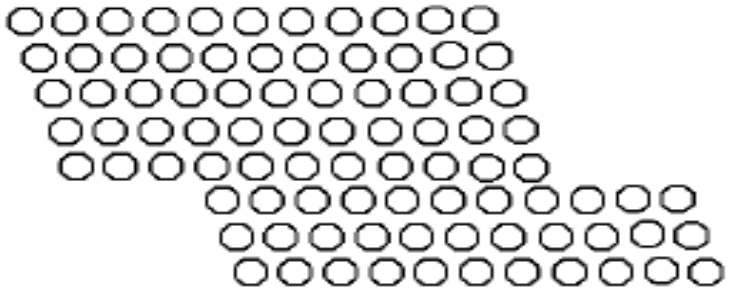
Twinning

- It results when a portion of crystal takes up an orientation that is related to the orientation of the rest of the untwined lattice in a definite, symmetrical way.
- The important role of twinning in plastic deformation is that it causes changes in plane orientation so that further slip can occur.
- Twinning also occurs in a definite direction on a specific plane for each crystal structure.

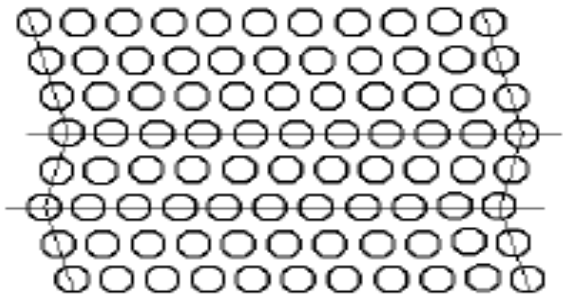
Slip Vs. Twinning



Undeformed Crystal



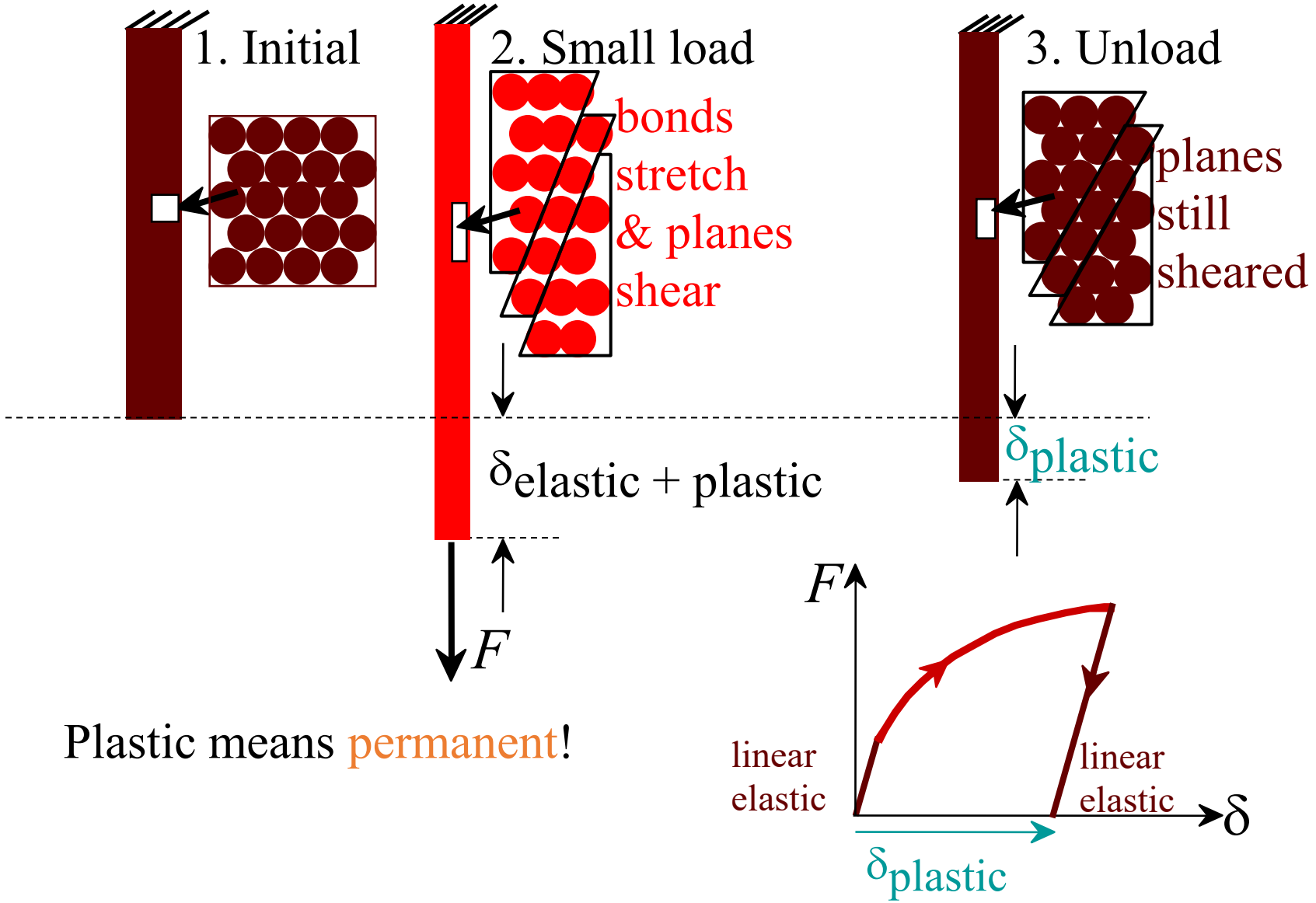
After Slip



After Twinning

	during/in slip	**during/in twinning**
Crystal orientation	Same above and below the slip plane	Differ across the twin plane
Size (in terms of inter-atomic distance)	Multiples	Fractions
Occurs on	Widely spread planes	Every plane of region involved
Time required	Milli seconds	Micro seconds
Occurrence	On many slip systems simultaneously	On a particular plane for each crystal

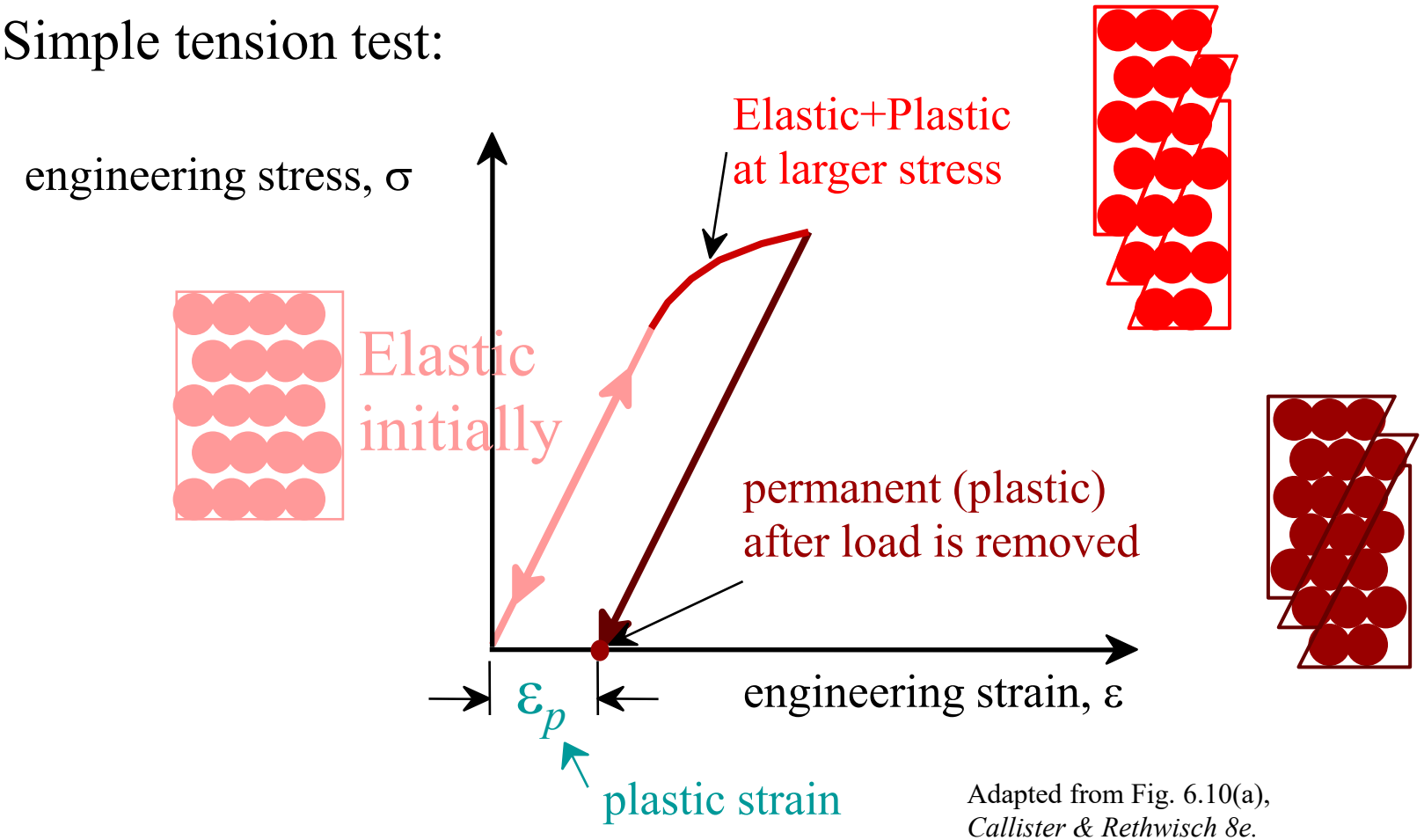
Plastic Deformation (Metals)



Plastic means permanent!

Plastic (Permanent) Deformation (at lower temperatures, i.e. $T < T_{melt}/3$)

- Simple tension test:



Adapted from Fig. 6.10(a), *Callister & Rethwisch 8e.*

Mechanical Behaviour

- Initially, strain is proportional to stress and shows reversible behaviour.
- i.e. When stress is removed the strain disappears.
- The modulus of elasticity (Young's Modulus) is the ratio between the stress σ and the reversible strain ε.

$$E = \sigma/\varepsilon$$

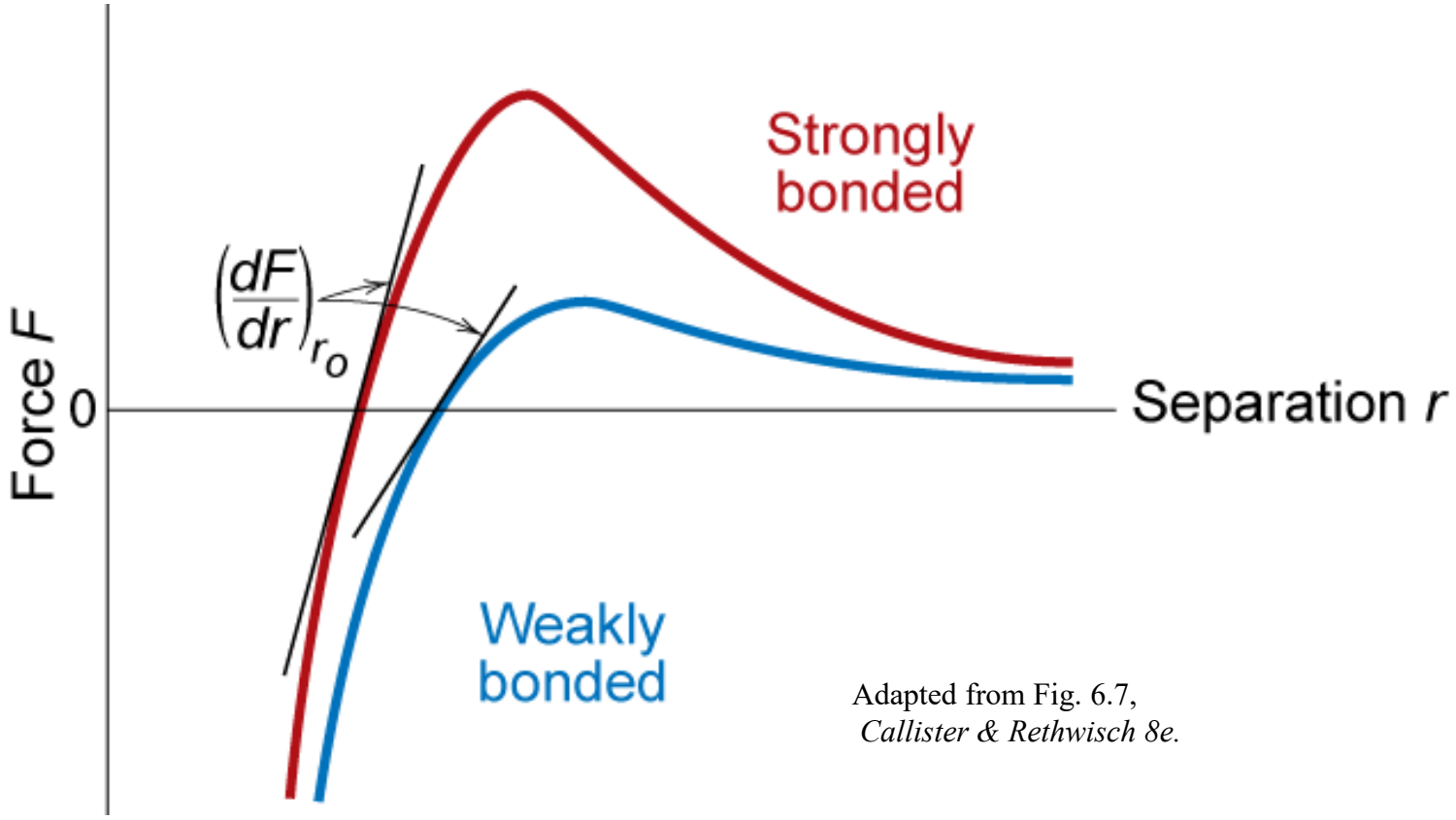
- The units of young's modulus is psi or pascals.

- The value of elastic modulus is a measure of the interatomic bonding forces and relates to the rigidity of engineering designs.

- *at higher stresses, permanent displacement can occur among the atoms within a material. In this case much of the strain is not reversible when the applied stresses are removed. In contrast with previous (elastic) strain this is called plastic strain.*

Mechanical Properties

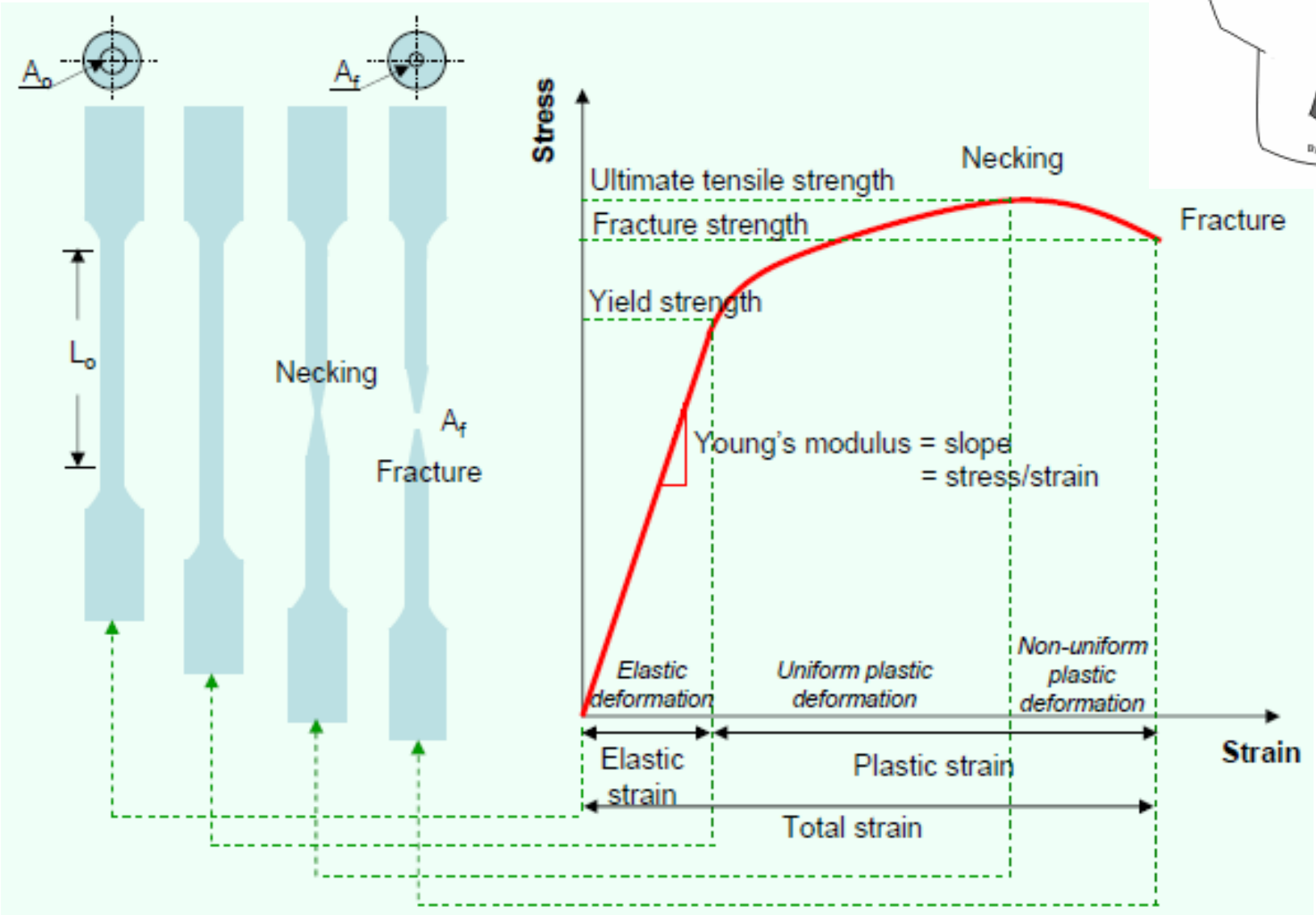
- Slope of stress-strain plot (which is proportional to the elastic modulus) depends on bond strength of metal



Adapted from Fig. 6.7, *Callister & Rethwisch 8e.*

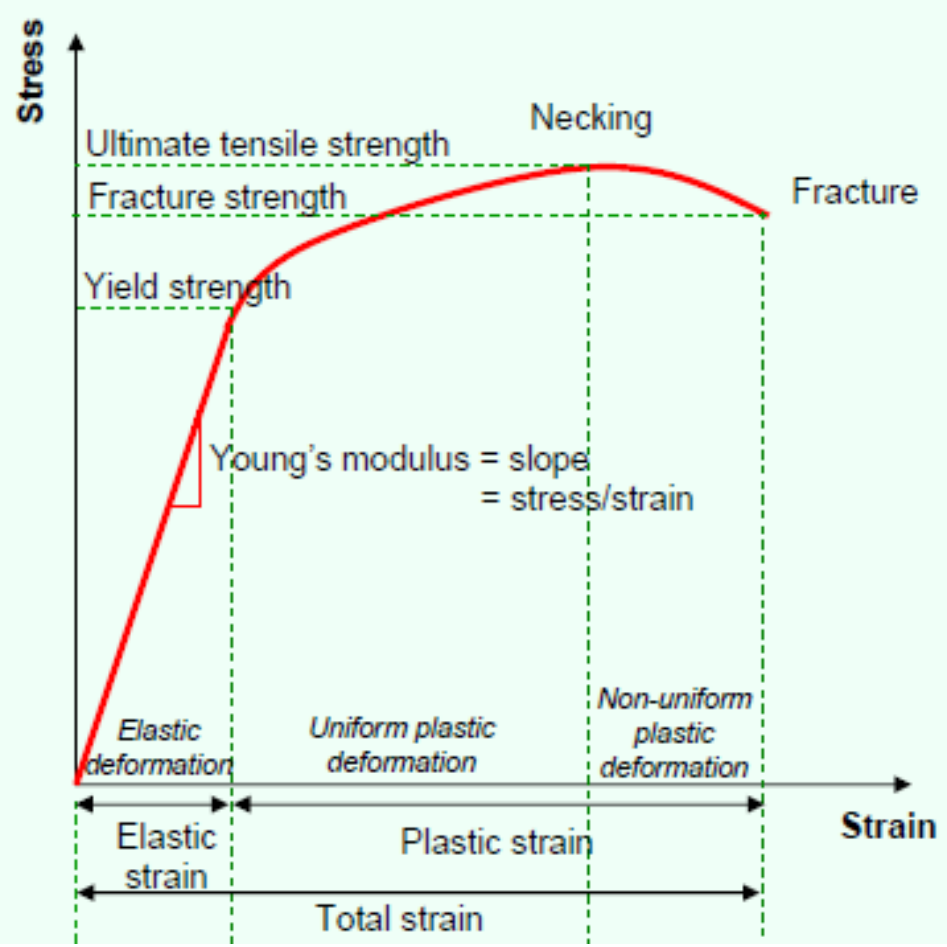
Engineering stress-strain curve





Engineering stress-strain curve

- Basic design information on the strength of materials.
- An acceptance test for the specification of materials.



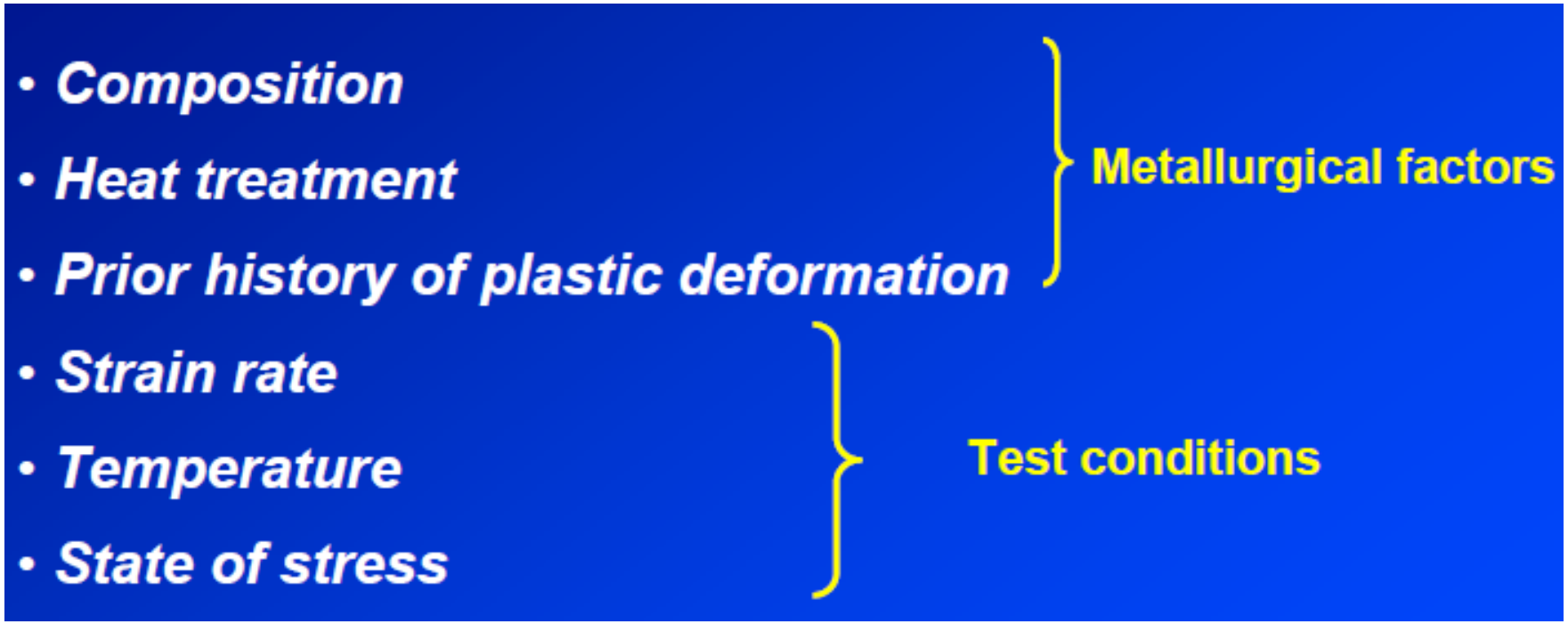
Longitudinal tensile stress

$$\sigma = \frac{P}{A_o}$$

linear strain

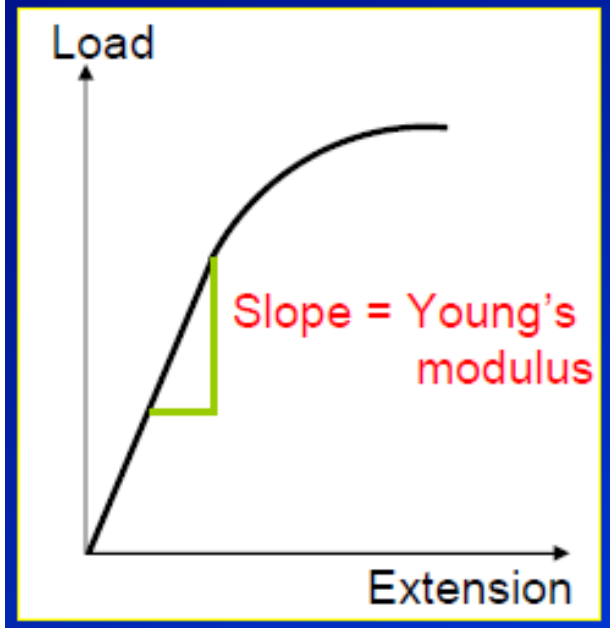
$$\varepsilon = \frac{\delta}{L_o} = \frac{\Delta L}{L_o} = \frac{L - L_o}{L_o}$$

Factors affecting shape and magnitude of stress-strain curve



Modulus of elasticity

Modulus of elasticity or ***Young's modulus*** is a measure of material stiffness (given by the slope of the stress-strain curve).



- Modulus of elasticity is determined by the binding forces between atoms (structure insensitive property)
- Cannot change ***E,*** but can improve by forming composites.
- Only slightly affected by alloying addition, heat treatment or cold work.

Temp ↑ ***Young's modulus*** ↓

Young's modulus ↑

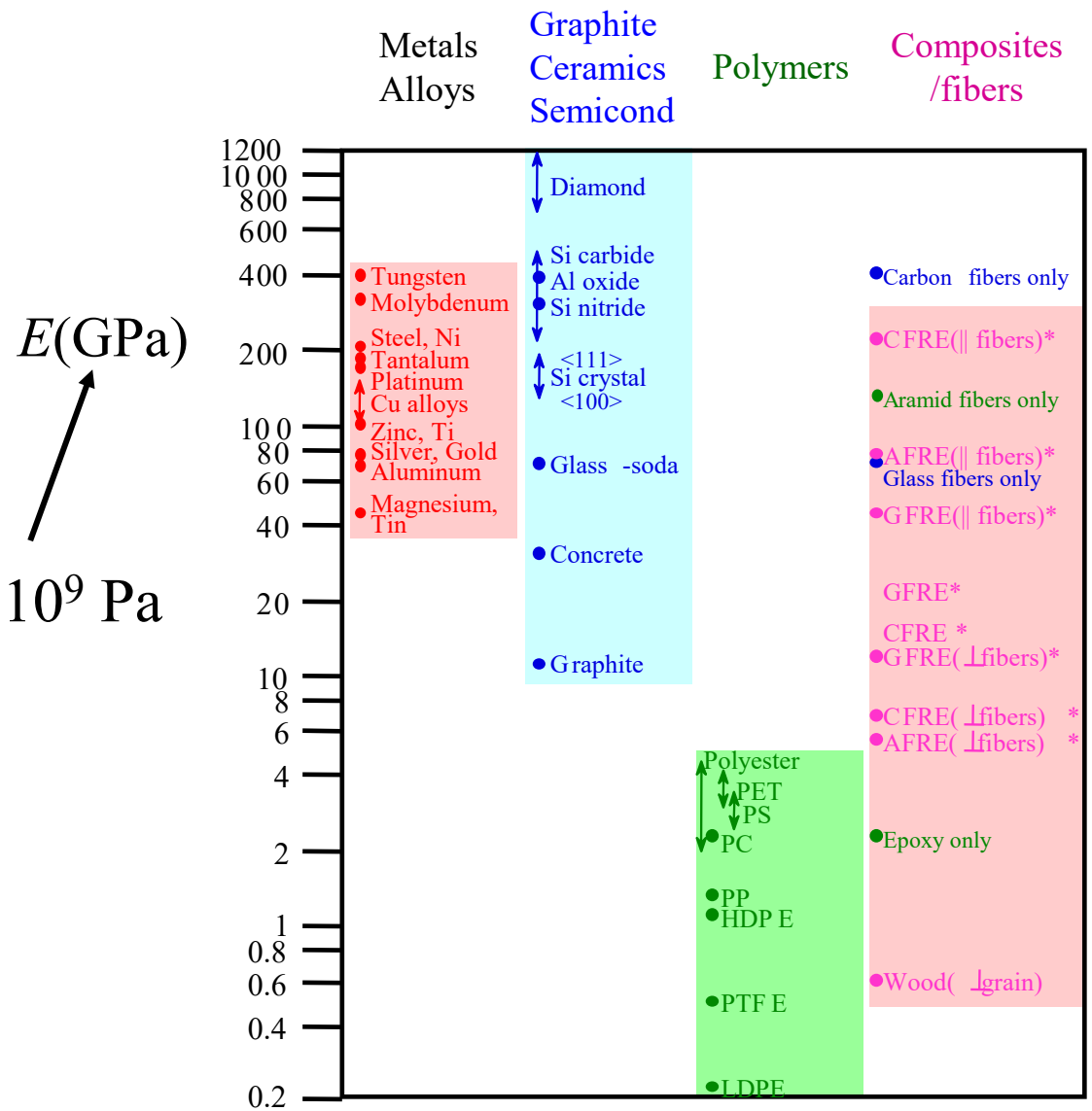
Stiffness ↑

Deflection ↓

Table 8-1 Typical values of modulus of elasticity at different temperatures

	Modulus of elasticity, GPa				
Material	Room temp.	477 K	700 K	810 K	922 K
Carbon steel	207	186	155	134	124
Austenitic stainless steel	193	176	159	155	145
Titanium alloys	114	97	74	70	
Aluminum alloys	72	66	54		

Young's Moduli: Comparison



Based on data in Table B.2, *Callister & Rethwisch 8e.*
Composite data based on reinforced epoxy with 60 vol% of aligned carbon (CFRE), aramid (AFRE), or glass (GFRE) fibers.

Yielding

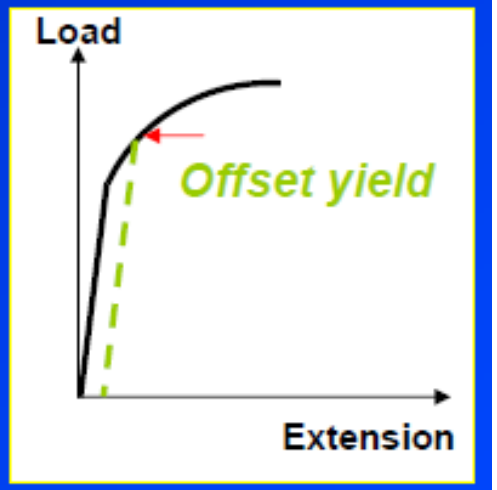
*Various criteria for the **initiation of yielding** are used depending on the sensitivity of the strain measurements and the intended use of the data.*

1) ***True elastic limit***: based on microstrain measurement at strains on order of 2×10^{-6}. Very low value and is related to the motion of a few hundred dislocations.

2) ***Proportional limit:*** the highest stress at which stress is directly proportional to strain.

3) ***Elastic limit:*** is the greatest stress the material can withstand without any measurable permanent strain after unloading. ***Elastic limit > proportional limit.***

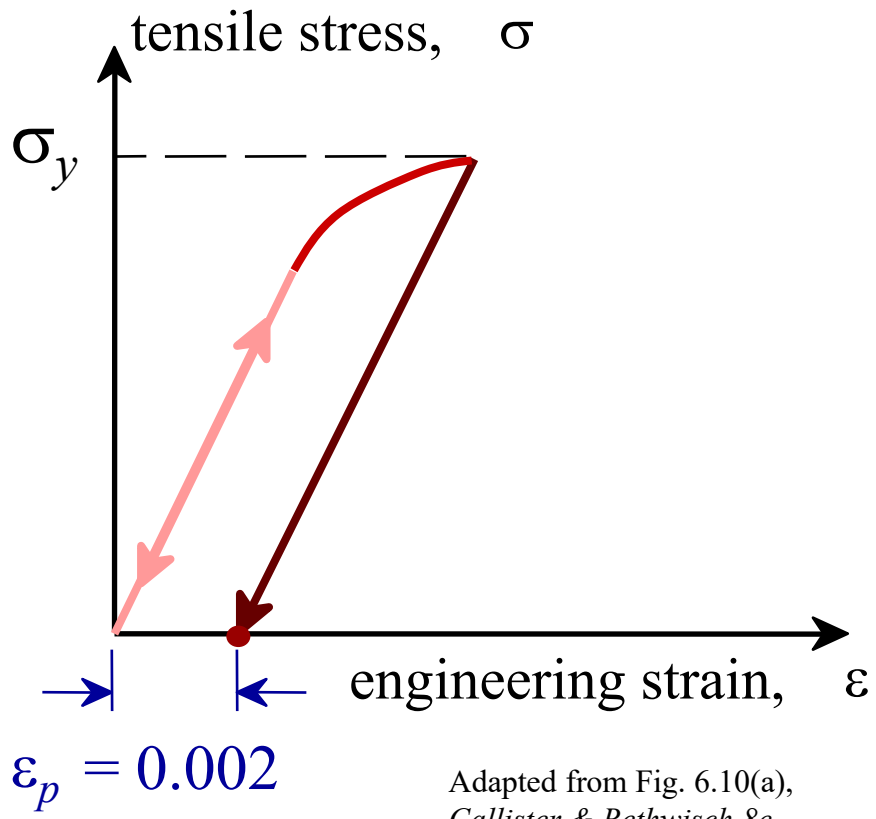
4) ***Yield strength*** is the stress required to produce a small specific amount of deformation.



Yield Strength, σ_y

- **Stress at which *noticeable* plastic deformation has occurred.**

when $\varepsilon_p = 0.002$



Adapted from Fig. 6.10(a), *Callister & Rethwisch 8e.*

σ_y = yield strength

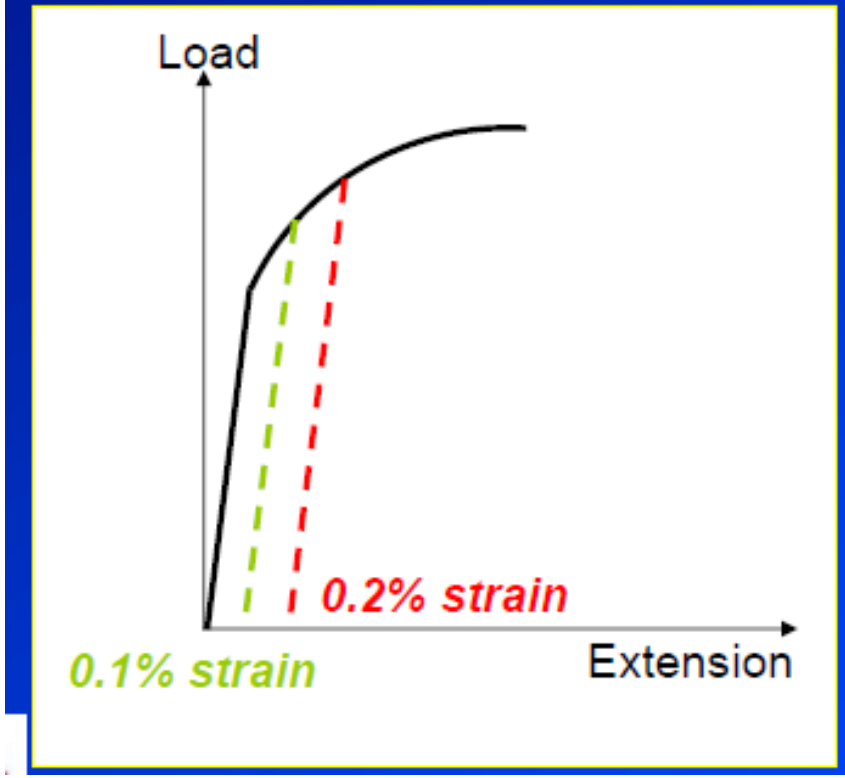
Note: for 2 inch sample

$\varepsilon = 0.002 = \Delta z/z$

$\therefore\ \Delta z = 0.004$ in

Yield strength of materials

The offset yield strength can be determined by the stress corresponding to the intersection of the stress-strain curve and a line parallel to the elastic line offset by a strain of 0.2 or 0.1%. (e = 0.002 or 0.001)



$$s_o = \frac{P_{(strain\ offset=0.002)}}{A_o}$$

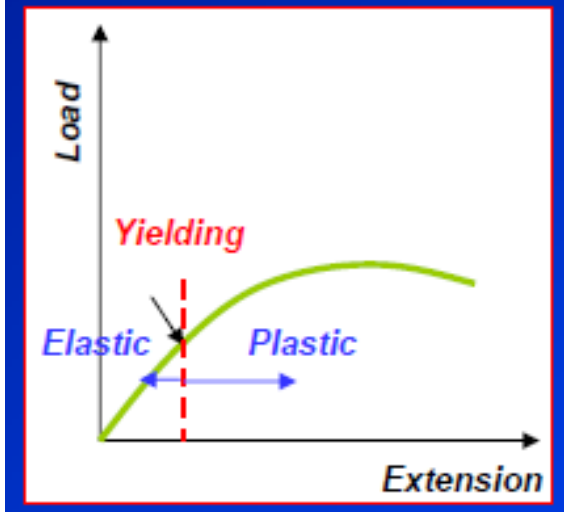
Eq.5

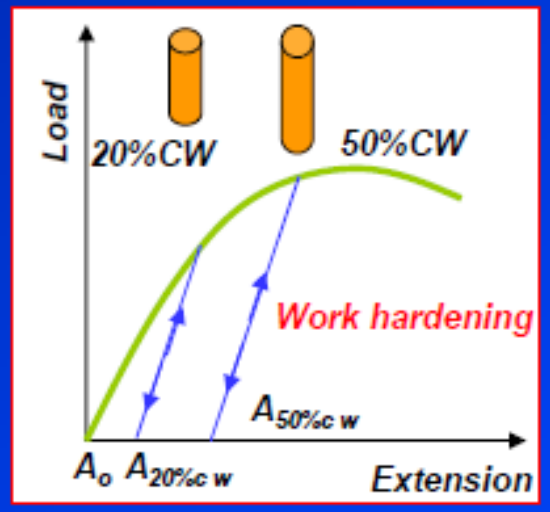
In Great Britain, the offset yield stress is referred to ***proof stress*** either at 0.1 or 0.5%strain.

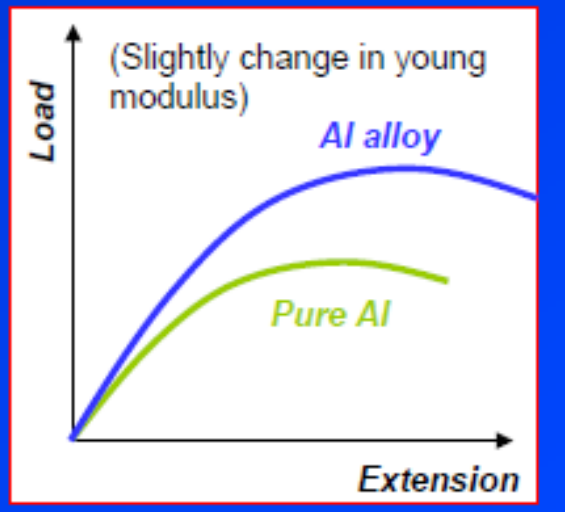
Used for design and specification purposes to avoid the practical difficulties of measuring the elastic limit or proportional limit.

Yield strength of materials

- ***FCC*** lattice materials (***Al, Cu***) have no definite yield point. The yield strength is therefore defined by the ***offset of yielding***.
- ***Yield strength*** can be improved by ***work hardening*** (cold working). → up to 300:1 stronger than original.
- Alloying of ***Al*** can improve ***elastic limit*** 1.5-2 times.







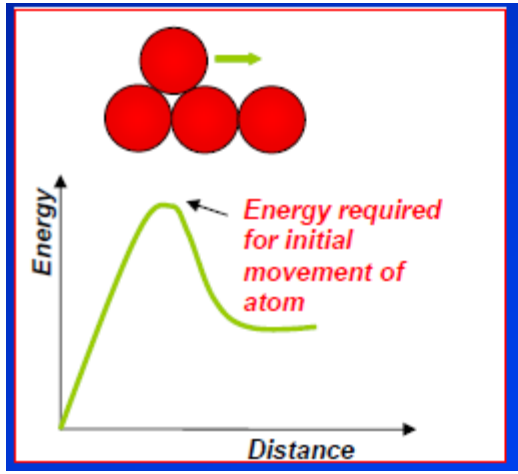
Improvement of yielding by cold working

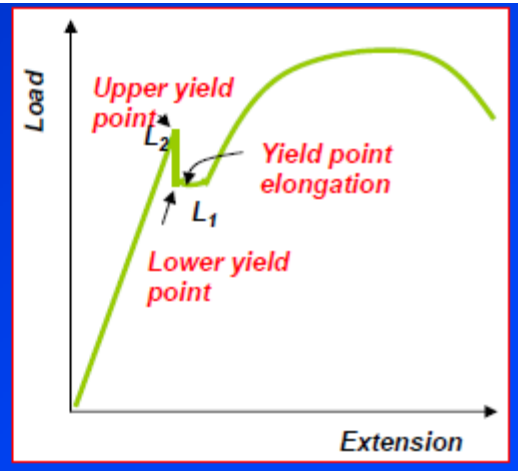
Improvement in elastic limit by alloying

Yield strength of materials

- **BCC** lattice materials (**Fe**) show a yield point phenomenon → ***Upper and lower yield points*** (depending on testing machine).
- ***Condition:*** Polycrystalline & small amounts of interstitial solute atoms.

Upper yield point $\frac{L_2}{A_o}$ **Lower yield point** $\frac{L_1}{A_o}$





At yield point, localised internal friction requires more energy for interstitial atom to move dislocation, after that dislocation are free from interstitial atom (carbon, nitrogen).

Ductility & determination

Ductility

Ductility is another important mechanical property. It is a measure of the degree of plastic deformation that has been sustained at fracture. A metal that experiences very little or no plastic deformation upon fracture is termed *brittle*. The tensile stress–strain behaviors for both ductile and brittle metals are schematically illustrated in Figure 6.13.

Ductility may be expressed quantitatively as either *percent elongation* or *percent reduction in area*. The percent elongation %EL is the percentage of plastic strain at fracture, or

t

$$\%\text{EL} = \left(\frac{l_f - l_0}{l_0}\right) \times 100 \qquad (6.11)$$

Ductile and brittle behaviour

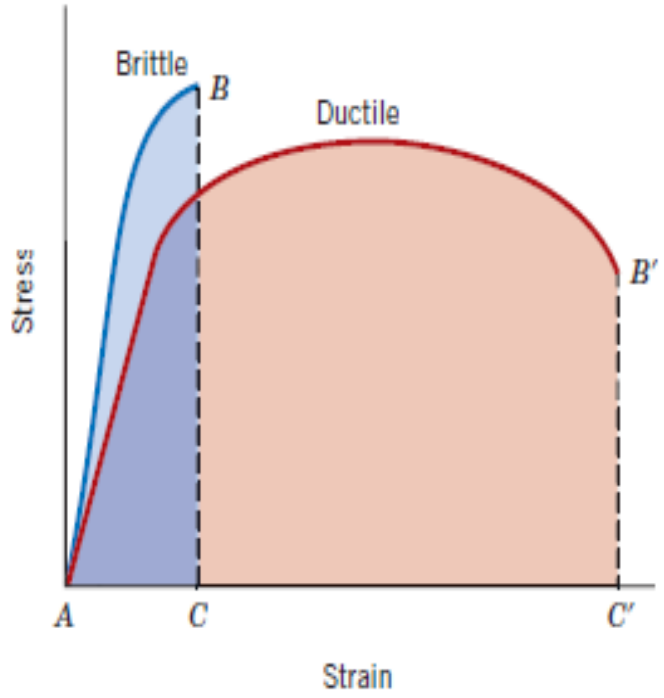


Figure 6.13 Schematic representations of tensile stress–strain behavior for brittle and ductile metals loaded to fracture.

Determination of Ductility

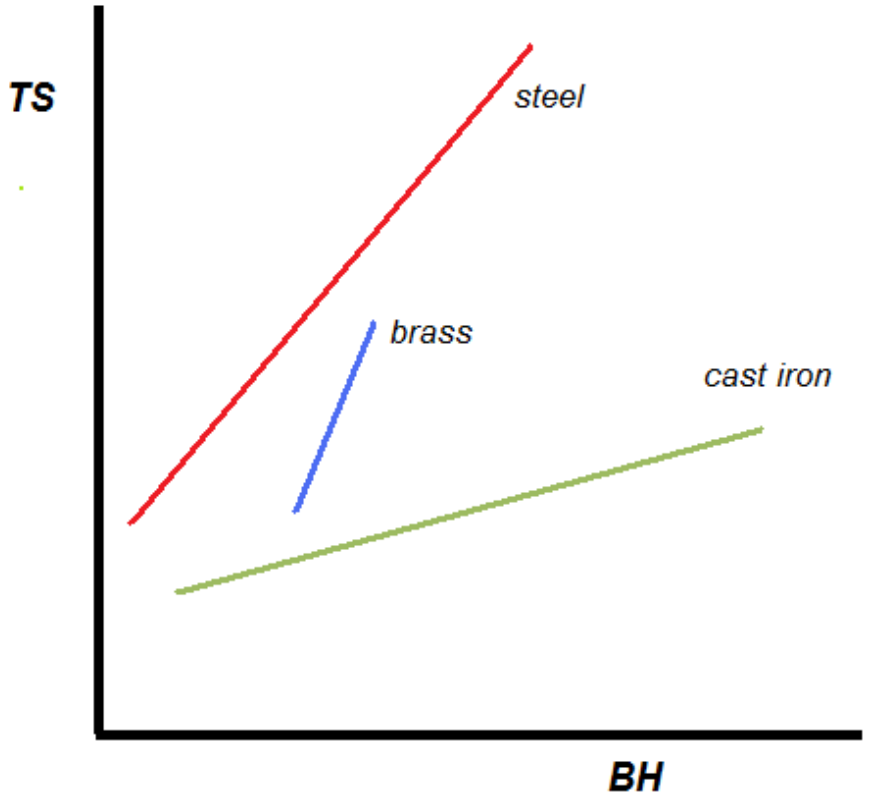
Ductility, as percent reduction in area

$$\%RA = \left(\frac{A_0 - A_f}{A_0} \right) \times 100 \qquad (6.12)$$

- ***Strength (and Hardness).***
- *The ability of a material to resist plastic deformation is called the yield strength (ys).*
- *This is calculated by dividing the force required to initiate the yield (first plastic deformation) by the cross-sectional area.*
- *In mild steel the yield strength is marked by a definite yield point is not well defined, the yield strength is defined as that stress required to produce 0.2% plastic offset.*

- ***Hardness*** *is the resistance of a material to penetration of its surface. The hardness and strength are related so that for a wide range of metallic materials strength increases with hardness.*



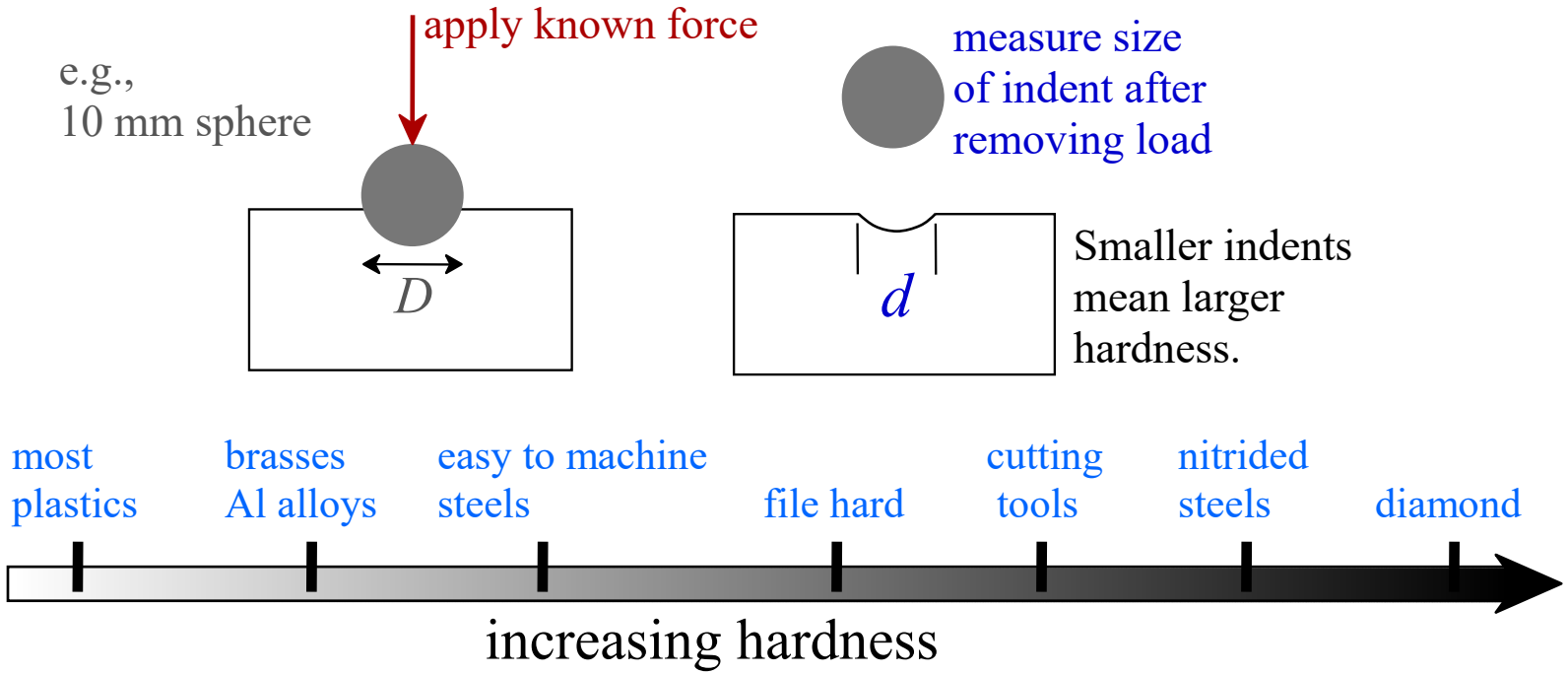






Hardness

- Resistance to permanently indenting the surface.
- Large hardness means:
  -- resistance to plastic deformation or cracking in compression.
  -- better wear properties.



Hardenability

- It is the ability of a material to attain the hardness by heat treatment processing. It is determined by the depth up to which the material becomes hard.

- The SI unit of Hardenability is meter (similar to length).

- Hardenability of material is inversely proportional to the weld-ability of material

Fatigue

❖ Fatigue is the weakening of material caused by the repeated loading of the material.

❖When a material is subjected to cyclic loading, and loading greater than certain threshold value but much below the strength of material (ultimate tensile strength limit or yield stress limit), microscopic cracks begin to form at grain boundaries and interfaces. Eventually the crack reaches to a critical size.

❖This crack propagates suddenly and the structure gets fractured. The shape of structure affects the fatigue very much. Square holes and sharp corners lead to elevated stresses where the fatigue crack initiates.

Creep

- Creep is the property of a material which indicates the tendency of material to move slowly and deform permanently under the influence of external mechanical stress.

- It results due to long time exposure to large external mechanical stress with in limit of yielding

- Creep is more severe in material that are subjected to heat for long time

STRENGTHENING MECHANISMS

- This has to do with the increase in yield stress
- Yielding (plastic deformation) is due to the movement of dislocation within the lattice of the material
- The principle of strengthening mechanism is increased obstacles to dislocation motion
- Strengthening Methods include:

(a) work (strain) hardening (single phase)

(b) grain refinement or grain-boundary strengthening (single phase)

(c) solid-solution strengthening (single phase)

(d) Quenched hardening (alloys)

(e) Precipitation hardening (alloys)

Strengthening mechanisms

Introduction

- *Solid Solution strengthening, grain size refinement, and strain hardening* (cold work) are three mechanisms by which single phase metal alloys may be strengthened and hardened
- These mechanisms may be used in conjunction with one another: e.g. solid solution strengthened alloy may also be strain hardened
- strengthening by grain size reduction and strain hardening may be eliminated or reduced by elevated temperature heat treatment

1. Strain-hardening (work-hardening)

- Strain-hardening is the phenomenon whereby a ductile metal becomes *harder* and *stronger* as it is *plastically deformed*.
- Also called work hardening because it occurs at low temps relative to the absolute temp. of the metal. (cold working temps)
- Degree of plastic deformation is expressed as *per cent cold work* (%CW) as:

$$\%CW = (A_o - A_d)/A_o * 100$$

where A_o is original area of cross section

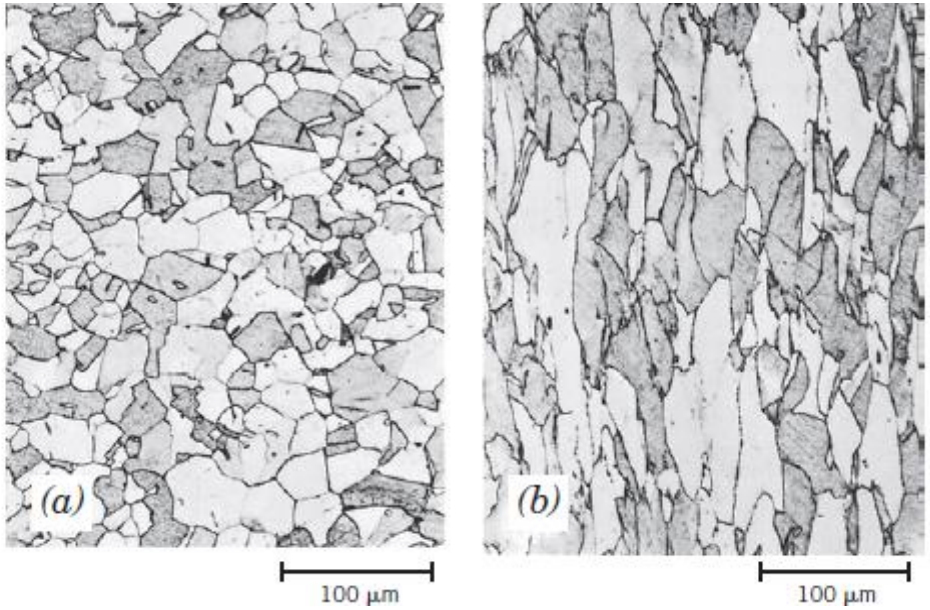
A_d is cross sectional area after deformation

Cold working (or strain hardening)

- It is the severe deformation of a metal at ambient temps., e.g. by cold rolling or drawing
- It causes a dense array of dislocations and disorders the crystal structure
- The dense dislocations interact and act as obstacles to further deformation
- It results in an increase in *yield strength* and *hardness*, but decrease in *ductility* and *toughness*

Effect of Strain-hardening (cold working) on microstrcuture

Figure 7.11 Alteration of the grain structure of a polycrystalline metal as a result of plastic deformation. (*a*) Before deformation the grains are equiaxed. (*b*) The deformation has produced elongated grains. 170×. (From W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, p. 140. Copyright © 1964 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.)



Effect of Cold work on stress-strain behaviour

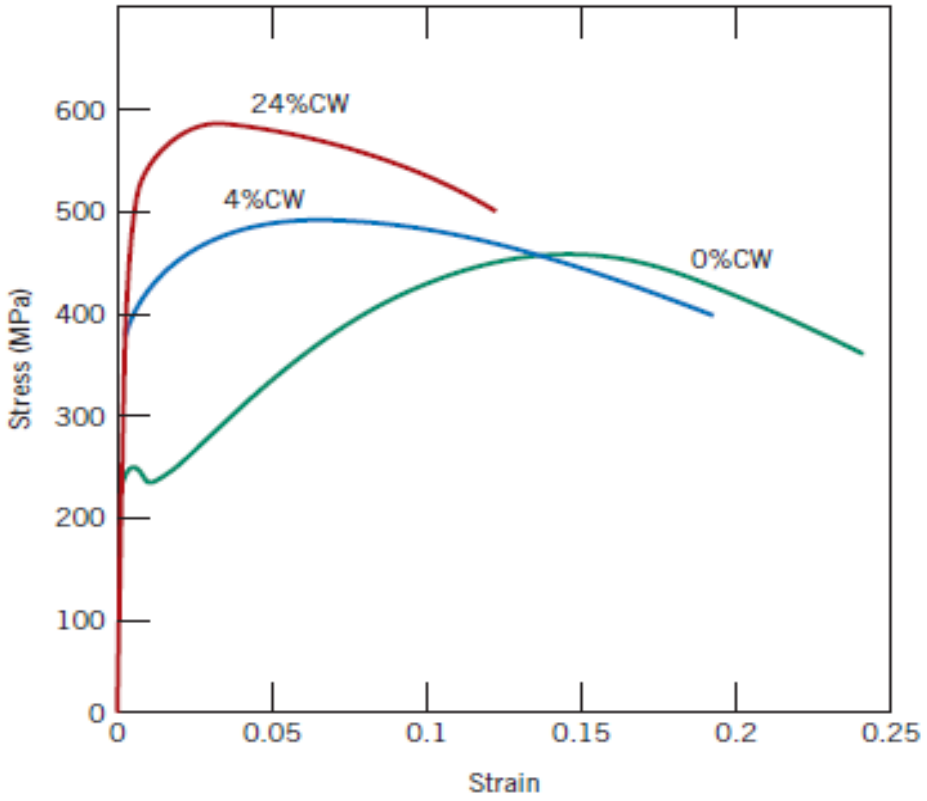


Figure 7.20 The influence of cold work on the stress–strain behavior of a low-carbon steel; curves are shown for 0%CW, 4%CW, and 24%CW.

2. Grain refinement (or grain- boundary strengthening)

- Grain boundary (g. b) serve as an effective barrier to movement of *glide dislocation*
- At low temps. Grain boundaries are stronger than grains; at high temperatures, grains are stronger than grain boundaries;
- Hall-Petch eqn: $\sigma_y = \sigma_i + k/(\sqrt{d})$ (.....)

σ_y = yield strength of poly-crystalline material

σ_i = overall resistance of lattice to dislocation movement

k = “locking parameter” of g.b.

d= grain size

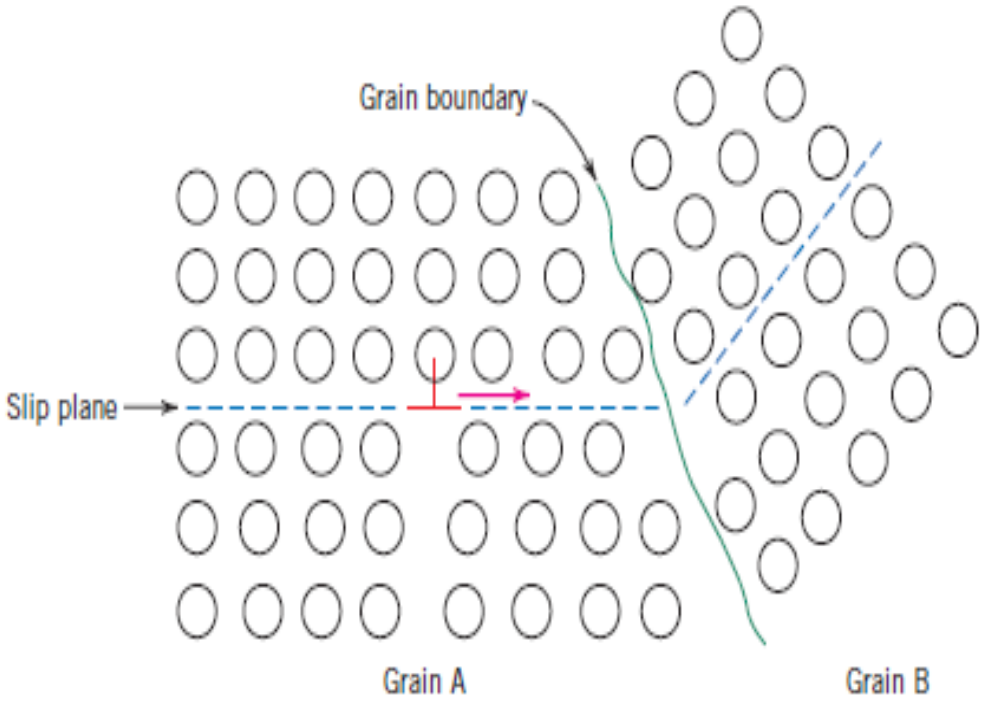
Grain Refinement

The size of the grains, or average grain diameter, in a polycrystalline metal influences the mechanical properties. Adjacent grains normally have different crystallographic orientations and, of course, a common grain boundary, as indicated in Figure 7.14. During plastic deformation, slip or dislocation motion must take place across this common boundary—say, from grain A to grain B in Figure 7.14. The grain boundary acts as a barrier to dislocation motion for two reasons:

1. Because the two grains are of different orientations, a dislocation passing into grain B will have to change its direction of motion; this becomes more difficult as the crystallographic misorientation increases.
2. The atomic disorder within a grain boundary region will result in a discontinuity of slip planes from one grain into the other.

Grain-Refinement

Figure 7.14 The motion of a dislocation as it encounters a grain boundary, illustrating how the boundary acts as a barrier to continued slip. Slip planes are discontinuous and change directions across the boundary. (From *VAN VLACK, TEXTBOOK MATERIALS TECHNOLOGY, 1st, © 1973.* Electronically reproduced by permission of Pearson Education, Inc., Upper Saddle River, New Jersey.)



Grain-size refinement

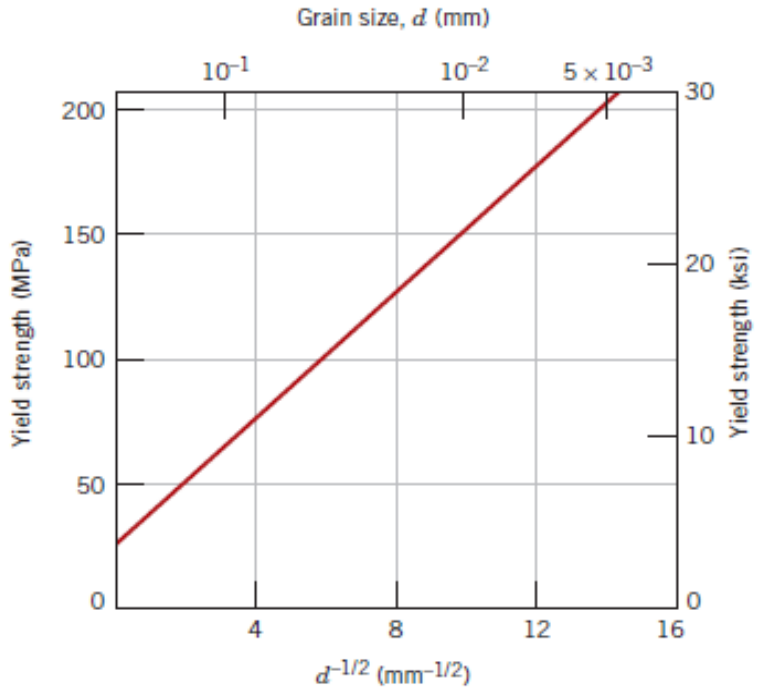


Figure 7.15 The influence of grain size on the yield strength of a 70 Cu–30 Zn brass alloy. Note that the grain diameter increases from right to left and is not linear. (Adapted from H. Suzuki, "The Relation between the Structure and Mechanical Properties of Metals," Vol. II, *National Physical Laboratory, Symposium No. 15,* 1963, p. 524.)

Hall–Petch equation—dependence of yield strength on grain size

$$\sigma_y = \sigma_0 + k_y d^{-1/2} \tag{7.7}$$

Grain-size refinement

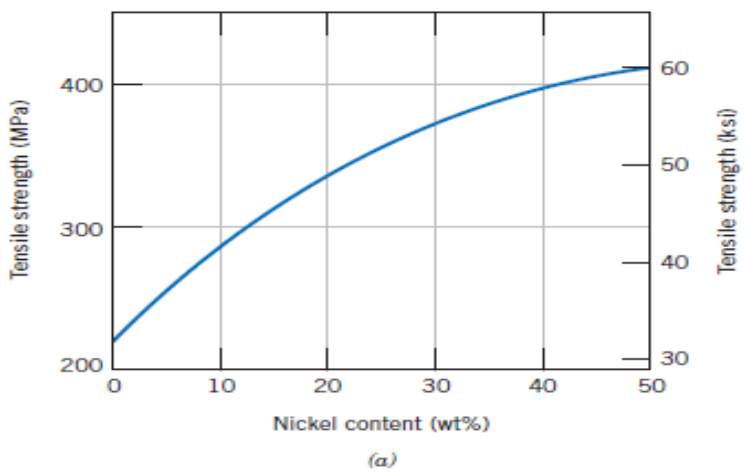
- Grain size refinement may be achieved by:

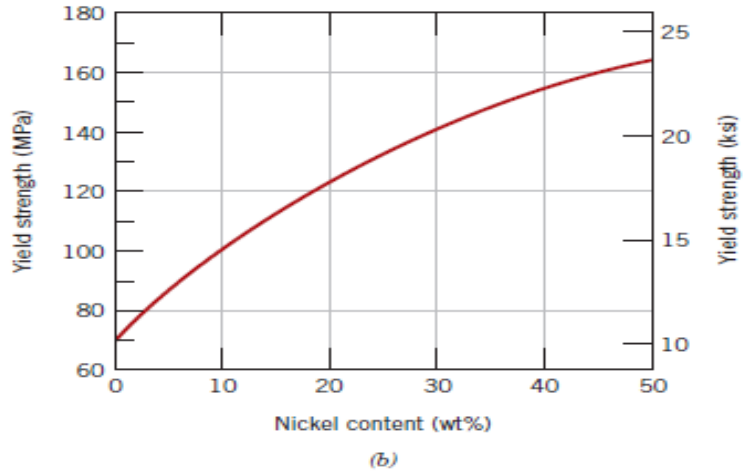
1. Rate of solidification. The higher the rate, the finer the grain size
2. Plastic deformation, followed by appropriate heat treatment (e.g. annealing)
3. grain size refinement, not only improves strength, but toughness as well

Solid-solution strengthening

- Pure metals are softer and weaker than their alloys
- Impurity atoms form interstitial or substitutional solid solution
- Increase in impurity atom concentrations increases yield strength, ultimate tensile strength and hardness
- Impurity atoms impose lattice strains on the surrounding host atoms
- Lattice strains interactions results between dislocations and the impurity atoms, restricting/hindering motion of dislocations
- Impurity atom smaller than host atom it substitutes exerts tensile strains on surrounding atoms
- A larger substitutional atom impinges compressive strains in the vicinity

Solid solution strengthening contd.





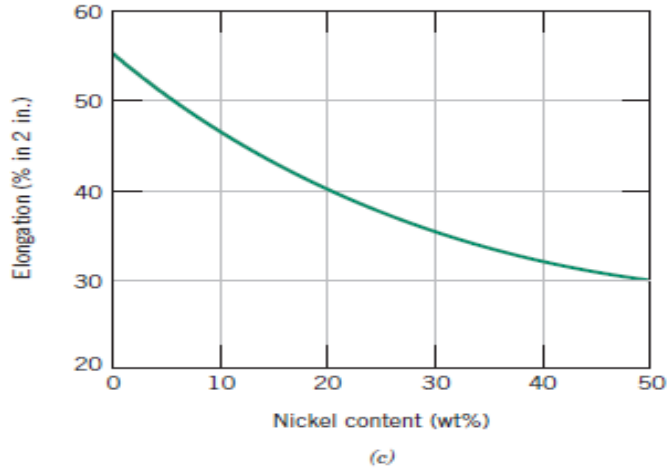


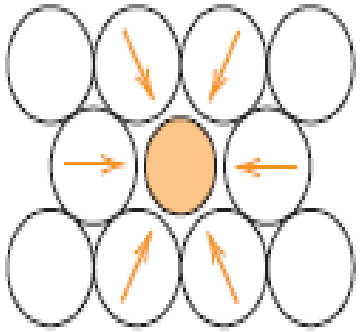
Figure 7.16 Variation with nickel content of (*a*) tensile strength, (*b*) yield strength, and (*c*) ductility (%EL) for copper–nickel alloys, showing strengthening.

Solid solution strengthening (contd.)

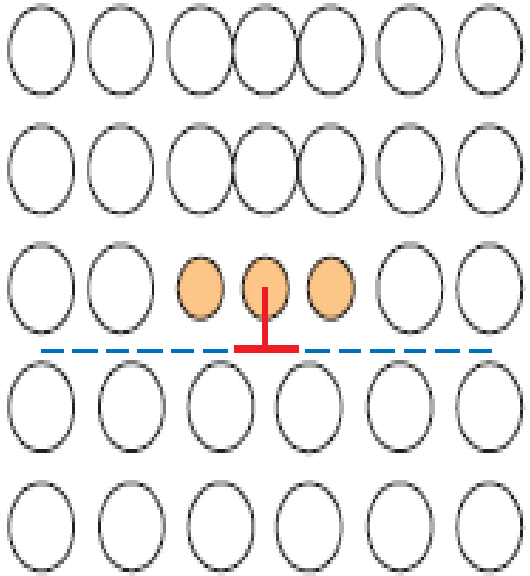
- The solute atoms tend to diffuse and segregate around dislocations….
- They arrange themselves to reduce the overall strain energy…cancel some of the strain in the lattice surrounding the dislocations
- How? … smaller impurity atom is located where its tensile strain will partially nullify some of the dislocation’s compressive strain (Fig.XX (a))
- A larger impurity atom will do the opposite… locate where its compressive strain will partially nullify some of the dislocation’s tensile strains (Fig.(b))

Solid solution strengthening (contd.)

Figure 7.17 (*a*) Representation of tensile lattice strains imposed on host atoms by a smaller substitutional impurity atom. (*b*) Possible locations of smaller impurity atoms relative to an edge dislocation such that there is partial cancellation of impurity–dislocation lattice strains.



(*a*)



(*b*)

Solid solution strengthening (contd.)

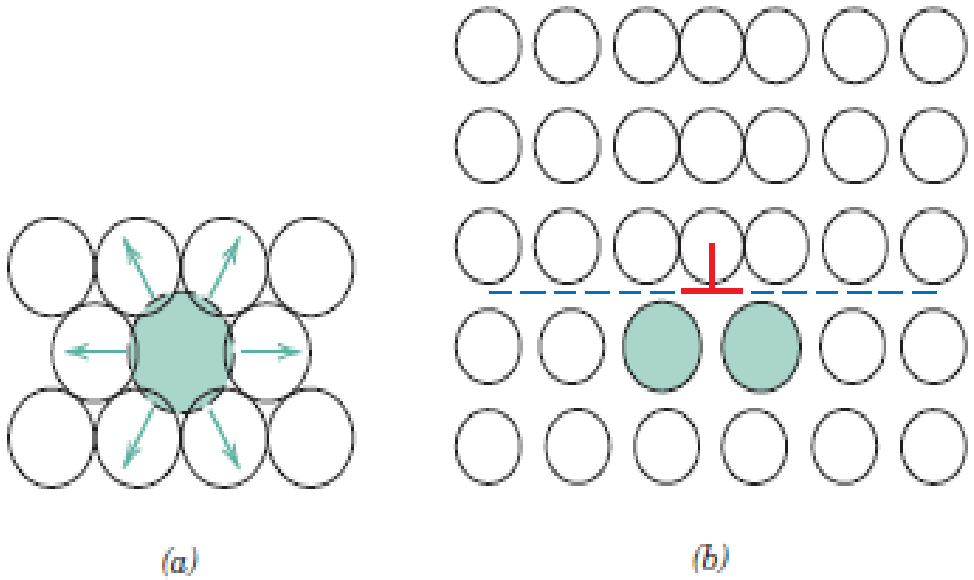


Figure 7.18 (*a*) Representation of compressive strains imposed on host atoms by a larger substitutional impurity atom. (*b*) Possible locations of larger impurity atoms relative to an edge dislocation such that there is partial cancellation of impurity–dislocation lattice strains.